Dr. D. Y. Patil Educational Federation's

Dr. D. Y. Patil College of Engineering & Innovation

Approved by AICTE, Recognized by Govt. of Maharashtra

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Accredited by NAAC with "A" Grade

Department of Artificial Intelligence and Machine Learning

Program Name: S.Y. B. Tech Class: S.Y. B. Tech Division: A

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SMARTBRIDGE | SKILLWALLET

AI Loan Eligibility Checker — BFSI Financial Intelligence Platform

PROJECT DESCRIPTION :

AI Loan Eligibility Checker is a web-based financial intelligence platform designed to help users evaluate loan affordability and understand factors that influence lending decisions. Users can enter information such as monthly income, existing debt obligations, employment status, requested loan amount, loan tenure and loan purpose. The system calculates financial indicators and presents an eligibility assessment with an estimated borrowing limit.

The platform combines a responsive web interface, a Python Flask backend, financial calculation logic and optional AI-powered guidance. It includes a loan eligibility engine, credit score analyzer, EMI calculator and an interactive financial-advisory assistant. The project is intended as a decision-support and educational tool; its estimates do not guarantee loan approval and do not replace a lender's formal assessment.

TECHNOLOGY & AI :

- Backend / Web Framework: Python 3.14 / Flask.
- Frontend: HTML5, CSS3 and modular JavaScript (ES6+)
- AI Integration: Anthropic Claude API (optional, with heuristic fallback)
- Financial Calculations: DTI, FOIR and annuity-based EMI calculations.
- Data Persistence: JSON file ledger and optional Google Sheets synchronization.
- Development Environment: Visual Studio Code / Python environment.

KET FEATURES :

- Loan eligibility assessment using applicant financial details.
- Debt-to-Income (DTI) and Fixed Obligation to Income Ratio (FOIR) calculations.
- Maximum affordable monthly EMI and estimated eligible loan amount.
- Credit score analyzer with rating bands and factor breakdown.
- Interactive EMI calculator with principal-interest breakdown and amortization schedule.
- Prepayment simulator to estimate interest savings and reduced loan tenure.
- Claude AI financial tips, underwriting explanations and conversational assistance.

SCENARIOS :

Scenario 1: An applicant enters monthly income, existing debt, employment status, requested loan amount and tenure. The eligibility engine calculates DTI/FOIR measures, estimates affordable EMI and displays an indicative underwriting verdict.

Scenario 2: A user reviews their credit score and the factors affecting it. The score analyzer presents rating bands and interactive scenarios to help explain how selected credit-related changes may affect the displayed estimate.

Scenario 3: A user wants to plan a loan before applying. They enter the principal, interest rate and tenure in the EMI calculator to view monthly instalments, total interest, amortization details and possible savings from extra payments.

SKILLS REQUIRED :

- Python programming and Flask
- HTML5, CSS3 and JavaScript
- Financial mathematics: DTI, FOIR, EMI and amortization
- REST API development and JSON data handling
- Anthropic Claude API integration and environment configuration
- Google Sheets / Apps Script integration (optional)

MENTOR :

Mentor: _____

PROJECT STATUS :

- Project modules: Loan Eligibility Engine, Credit Score Analyzer, EMI Calculator, AI Financial Tips and Submission Ledger
- The repository README describes the modules and setup; formal epic, story and task counts can be added from the project tracker if available.

TECHINICAL ARCHITECTURE :

The application uses a browser-based frontend built with HTML, CSS and JavaScript, which communicates with a Python Flask backend. Backend logic calculates eligibility indicators and EMI estimates, and exposes API endpoints for eligibility, AI tips, conversational questions and submission records. Anthropic Claude can generate financial explanations when configured; a built-in heuristic fallback provides guidance when the API is unavailable. Application submissions can be stored in a JSON ledger and optionally synchronized with Google Sheets.

HAEDWARE REQUIREMENTS :

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent recommended.
- RAM: Minimum 8 GB recommended.
- Storage: Sufficient free space for Python, dependencies and project files.
- Internet connectivity for Claude API and optional Google Sheets integration.

SOFTWARE REQUIREMENTS :

- Operating System: Windows, macOS or Linux.
- Web Browser: Recent version of Chrome, Firefox or Microsoft Edge.
- IDE / Code Editor: Visual Studio Code or equivalent.
- Python 3.10 or above (repository README notes testing with Python 3.14)
- Git for version control (recommended)

INSTRUCTIONS :

- Clone or download the project repository from GitHub.
- Install the required packages using: python -m pip install -r requirements.txt
- Configure ANTHROPIC_API_KEY and ANTHROPIC_MODEL in the .env file if AI guidance is required.
- Configure the Google Sheets webhook or credentials only if synchronization is required.
- Start the application by running: python app.py
- Open http://127.0.0.1:5000 in a web browser and enter the required loan details.
- Review eligibility calculations and AI-generated guidance as estimates, not a formal lending decision.
- Keep API keys and service-account credentials private; do not commit secrets or sensitive applicant data to a public repository.

MILESTONES :

Milestone 1 – Project Planning & Requirement Analysis

Define the objective of the loan eligibility checker, identify applicant inputs and specify the main modules and user workflow.

Milestone 2 – Interface & Financial Data Management

Build the web interface for applicant details, credit score analysis, EMI calculation and presentation of financial results.

Milestone 3 – Backend & Eligibility Engine

Implement Flask routes and financial logic for DTI, FOIR, estimated eligible loan amount, EMI and submission handling.

Milestone 4 – AI Integration & Financial Guidance

Connect the optional Claude API for underwriting explanations, financial tips and conversational Q&A, with a fallback when the service is unavailable.

Milestone 5 – Testing, Data Persistence & Deployment

Validate financial calculations and API responses, test error cases, protect credentials and verify JSON-ledger and optional Google Sheets persistence.

PRIVACY,SAFETY & LIMITATIONS :

The platform provides indicative estimates based on user-entered information and predefined calculations. AI-generated advice may be incomplete or inaccurate, and credit-score simulations are illustrative rather than official score updates. Do not use the tool as the sole basis for borrowing decisions. Protect API keys and service-account files, minimize sensitive data collection and verify all loan terms directly with the relevant lender.

CONCLUSION :

AI Loan Eligibility Checker combines financial calculations, a credit score analyzer, an EMI planning tool and optional AI assistance in a single web platform. It demonstrates how Python, Flask, browser technologies and external APIs can support clearer financial decision-making by presenting affordability indicators and understandable guidance.

EXPECTED OUTCOME:

The expected outcome is a web application that calculates indicative loan eligibility, estimates EMI and borrowing capacity, explains credit-related factors and provides practical financial guidance. Further improvements may include stronger authentication, expanded testing, secure database storage, additional explainability, accessibility enhancements and carefully validated integrations with trusted financial data sources.